MUHAMMAD HASEEB

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SUMMARY

Machine Learning Engineer with 3+ years of industry experience shipping production LLM, NLP, and agentic AI systems at scale. Built conversational AI with intent recognition and multi-agent orchestration, RAG pipelines, LoRA fine-tuned models, and MLOps infrastructure serving 40+ microservices at 99.7% uptime. Research background in open-source LLM evaluation and automated feedback generation; first author of 2 accepted papers at WWW '26 and BEA 2026 @ ACL.

SKILLS

- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, AutoGen, Hugging Face, LLMs, RAG, LoRA, Unsloth, vLLM, VLMs, Prompt Engineering, Agentic Workflows
- **MLOps:** Docker, Kubernetes, Helm, MLflow, Model Monitoring, CI/CD, GitHub Actions, Azure Databricks, Airflow, Grafana, Prometheus
- **Cloud & Data:** GCP, AWS, Azure, PostgreSQL, MySQL, MongoDB, Pandas, PySpark, Delta Lake, REST APIs
- **Programming:** Python, JavaScript, C++, FastAPI, Node.js, React

EXPERIENCE

Graduate Assistant (Research), Civil Engineering Dept, Texas Tech University March 2025 – Present

TxDOT-Funded Transportation Safety Research Project

- Conducting applied research on crash causation analysis using statistical and GenAI-integrated frameworks; built a multi-agent NLP chatbot with intent classification (local LLM) routing queries across four specialized pipelines: filter updates, chart explanation, data querying, and general conversation; manuscript in preparation.
- Engineered an agentic report generation system using LangChain and GenAI for automated PDF, PPT, and poster creation, integrated into a custom React analytics dashboard with interactive safety visualizations.
- Designed automated ETL pipelines processing 100K+ district crash records with agentic RAG for natural language querying, enabling interactive safety pattern visualizations and corridor risk identification across 5+ TxDOT districts.

DEVSINC, Islamabad, Pakistan Jan 2022 – Jan 2025

Senior Software Engineer

- Led AI/ML projects automating construction design requirements using NLP and Generative AI, improving vendor proposal accuracy by 80% while supervising a team of 3 engineers to deliver production solutions.
- Architected MLOps infrastructure managing Kubernetes clusters on GCP (40+ microservices), implementing monitoring with Prometheus and Grafana, achieving 99.7% uptime and reducing incident response time by 60%.
- Developed a conversational AI system using open-source LLMs with LoRA fine-tuning and RAG pipelines, incorporating intent recognition and multi-agent orchestration to route queries to specialized task agents, improving accuracy from 74% to 88% on disease trends and treatment datasets.
- Spearheaded Azure Databricks feature store adoption to eliminate redundant feature engineering across teams, enabling feature reuse and reducing model deployment time from 8 weeks to 4 weeks.
- Built production ML pipelines with automated data and model drift monitoring, champion-challenger validation, and scheduled retraining workflows, reducing deployment cycle time by 50%.

PUBLICATIONS

- **Haseeb, M., et al.** "Toward Cross-Domain Automated Feedback: A Comparative Evaluation of Open-Source Models across Diverse Student Assessment Types." 21st Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2026) @ ACL 2026. **(Accepted)**
- **Haseeb, M., Amjad, A. I., Amjad, M., & Sheng, V. S.** "From Code to Rubrics: A Multimodal Evaluation of LLM Systems for Automated Feedback Generation." Companion Proceedings of the ACM Web Conference 2026 (WWW Companion '26), Dubai, UAE. ACM. **(Accepted)**

RESEARCH & ACADEMIC ACTIVITIES

Graduate Research Assistant, Texas Tech University Mar 2025 – Present

Master's Thesis: AI-Driven Feedback and Evaluation Systems for Personalized Learning using LLMs

- Designing and evaluating agentic LLM architectures for automated student feedback generation across programming, math, and writing; building open-source, locally deployable tools as alternatives to proprietary APIs for under-resourced institutions.

Conference Poster Presentation, Texas A&M University Sept 2025

- Built and presented a clinical AI system combining Quantum ML (trained on MIMIC dataset) and a RAG pipeline over PubMed for evidence-based case retrieval, designed to flag high-vulnerability patients and surface relevant clinical literature; presented at the Texas First Quantum Summit, Texas A&M University.

PROJECTS

- **HealthQuery Intelligence:** Built a conversational AI system for natural language querying over a cloud-hosted medical dataset, enabling users to extract insights from open-ended clinical queries.
- **SmartCompliance:** Built an AI-powered system using Generative AI and NLP to compare proposals against compliance standards, cutting review time by 80% and improving accuracy.
- **TradeHub Platform:** Implemented microservices for a trading platform managing the complete trade flow from order-making to clearing houses, creating a cohesive and efficient trading environment within a GCP cluster.

EDUCATION

Texas Tech University, Lubbock, Texas | *Master's in Computer Science* | Jan 2025 – Jul 2026 | CGPA: 4.0/4.0
FAST NUCES, Islamabad, Pakistan | *Bachelor's in Computer Science* | Aug 2018 – May 2022 | CGPA: 3.40/4.0